

How to Bake a Cake – Beginner's Guide

Lesson 1: Introduction to Cake Baking

Introduction to Cake Baking - Study Notes

Key Concepts

1. **Basic Cake Classifications** – Sponge cake vs. butter (or “American”) cake.
2. **Essential Baking Tools** – Mixing bowls, measuring devices, pans, mixers, ovens, and thermometers.
3. **Food Safety & Hygiene** – Proper ingredient storage, cross contamination prevention, and oven safety.
4. **Fundamental Techniques** – Creaming, whisking, folding, and proper mixing times.
5. **Ingredient Functions** – How flour, fat, sugar, eggs, leavening agents, and liquids work together.

Summary

Baking a cake begins with understanding the two most common families of cakes: **sponge cakes** and **butter (or “American”) cakes**. Sponge cakes rely on air incorporated into eggs for leavening, producing a light, airy crumb with little or no added fat. Butter cakes, on the other hand, use creamed butter (or other fats) and chemical leaveners (baking powder/soda) to create a richer, tender texture.

Equally important as the recipe itself are the **tools** and **safety practices** that ensure consistent results and a safe kitchen. Accurate measuring, proper mixing equipment, and the right bakeware shape the batter, while good hygiene—such as washing hands, keeping raw eggs refrigerated, and avoiding cross contamination—prevents foodborne illness. Understanding each ingredient's role helps you troubleshoot problems and adapt recipes with confidence.

Important Points

- **Sponge vs. Butter Cake**
 - **Sponge**: eggs (whole, yolks, or whites) provide lift; minimal fat; often folded with flour.
 - **Butter**: butter (or oil) creamed with sugar creates air bubbles; chemical leaveners add rise.
- **Core Tools**
 - Digital kitchen scale (for weight accuracy)
 - Set of measuring cups & spoons (for quick volume checks)
 - Stand or hand mixer, rubber spatula, whisk, and fine mesh sieve
 - Light colored metal or silicone cake pans; parchment paper or silicone liners
 - Oven thermometer (most ovens run hot or cold)
- **Safety & Hygiene**
 - Store eggs, dairy, and perishable items at 40 /40 °F (4 °C).
 - Wash hands before and after handling raw eggs.

- Clean surfaces and utensils between tasks.
- Use oven mitts; never leave a pre heated oven unattended.
 - **Ingredient Functions**
- **Flour**: provides structure via gluten formation.
- **Fat**: tenderizes, adds flavor, and carries air when creamed.
- **Sugar**: sweetens, tenderizes, and helps with browning (Maillard).
- **Eggs**: bind, emulsify, and leaven (especially whites).
- **Leaveners**: baking powder/soda create CO₂, to expand the batter.
- **Liquid**: hydrates flour, dissolves sugar, and creates steam.
 - **Mixing Techniques**
- **Creaming**: beat butter + sugar until pale & fluffy (butter cake).
- **Whisking**: beat whole eggs or yolks to incorporate air (sponge).
- **Folding**: gently combine dry ingredients or beaten egg whites without deflating air.

Examples

Example 1 – Simple Sponge Cake (Victoria Sponge)

1. **Ingredients (by weight)**: 200 /g flour, 200 /g sugar, 200 /g butter (room temp), 4 large eggs, 2 /tsp baking powder, pinch of salt.
2. **Method**:
 - Beat eggs with sugar until the mixture triples in volume and is pale ("H /5 /min).
 - Sift flour, baking powder, and salt; fold into the egg sugar mixture in three additions.
 - Melt butter, cool slightly, and fold in gently.
 - Pour into two 20 /cm round pans, bake at 180 /°C (350 /°F) for 20 25 /min.
3. **Why it works**: The leavening comes mainly from the air trapped in the whisked eggs; the small amount of butter adds moisture without weighing the crumb down.

Example 2 – Classic Butter Cake (Yellow Cake)

1. **Ingredients (by weight)**: 250 /g all purpose flour, 250 /g granulated sugar, 225 /g unsalted butter (softened), 3 large eggs, 2 /tsp baking powder, 1 /tsp vanilla extract, 120 /ml milk.
2. **Method**:
 - Cream butter and sugar together until light and fluffy ("H /3 4 /min).
 - Add eggs one at a time, beating after each addition; mix in vanilla.
 - Sift flour and baking powder; add alternately with milk, beginning and ending with flour.
 - Divide batter into two 20 /cm pans, smooth tops, bake at 175 /°C (350 /°F) for 25 30 /min.
3. **Why it works**: Creaming incorporates air that expands during baking, while the butter provides richness and a tender crumb. The chemical leavener ensures a reliable rise even if the batter isn't heavily aerated.

Quick Review

1. **What is the primary leavening source in a traditional sponge cake?**
2. **Name two essential safety habits when handling raw eggs.**
3. **Why is it important to use an oven thermometer?**

4. **List three functions of sugar in cake batter.**

5. **When should you fold ingredients instead of stirring vigorously?**

Further Reading

- **Ingredient Science** – How gluten development affects cake texture.
- **Advanced Mixing Methods** – The “reverse creaming” technique and its impact on crumb.
- **Gluten Free & Vegan Cake Alternatives** – Substituting eggs, dairy, and flour.
- **Baking Temperature & Altitude Adjustments** – Modifying recipes for high altitude locations.
- **Decoration Basics** – Simple buttercream, ganache, and fruit fillings.

Happy baking! Ø&β,

